

Benchmark for Crack Re-Identification in Ticket Management Systems

Délcio Ferramenta, Cátia Loureiro, António Silva, Beatriz Miranda,
Paulo Brito, Nuno da Costa, Ricardo Rodrigues, Duarte Fernandes
DTx CoLAB / University of Minho, Campus de Azurém,
Guimarães, Portugal
delcio.ferramenta@dtxcolab.pt

Abstract—When using a ticket management system, it is important to decide if a newly reported defect matches an existing ticket or should be logged as a new one. If the same crack is reported more than once from different angles, the record history can become fragmented, and the assessment work may be repeated. We call this process crack re-identification (re-ID) and have introduced CRACKID. The frozen benchmark snapshot includes 140 photographs, 17 walls, and 212 identities, with 6 validation walls and 11 test walls, including 19 matchable test identities. We evaluate ten zero-shot crop-based matchers from four families using retrieval, thresholded pairwise decisions, and open-set identification with rejection. Retrieval performance varies, with Rank-1 scores ranging from 0.370 to 0.613. When pairwise decisions are calibrated using the validation set, all methods perform just above the best-constant baseline. The pairwise F_1 at the validation threshold $F_1^{\text{@}v}$ is between 0.331 and 0.338, compared to a floor of 0.305, and the oracle F_1^* ranges from 0.337 to 0.374. The results show that wall location has a bigger impact than crack appearance, based on context and crack-erasure controls. A geometric reference baseline, which aligns the wall texture with cracks removed and compares the projected masks, achieves a detection and identification rate (DIR) of 0.768 at a false alarm rate (FAR) of 0.1 on the 48

Index Terms—Annotation Quality, Benchmark, Building Maintenance, Crack Re-Identification, Geometric Alignment, Homography, Structural Inspection

I. INTRODUCTION

Ticket management systems arrange reports of physical defects into records so that they can be inspected, prioritized, interventions carried out and followed up on. Although each ticket should contain the history of a single incident, different users may take photographs of the same crack more than once, which results in duplicate tickets that disrupt the history and lead to redundant assessment work. Whenever a new observation is made it must either be linked to the existing ticket or a new ticket had to be created for damage that has not been recorded.

This is about the visual re-identification (re-ID) of photographic evidence in that (i) the same shape can be recognised even though there are variations in the angle of view, distance, field of view, illumination, or partial visibility and (ii) damage which has changed (by extending, widening or branching) must still be identified, together with the need to tell the difference between damage that has grown and new damage that has appeared. Mistakes in this process lead to duplicates or misattribution.

The existing methods for detecting and segmenting cracks are concentrated on dealing with individual images. Since datasets like CrackSeg9k [1] and OmniCrack30k [2] do not include cross-observation identities, it is not possible to associate observations with tickets. It has not yet been tested whether techniques from person/object re-ID [3], [4] can be applied to cracks—given that different defects may look similar to one another and that one defect can vary considerably.

We have developed CRACKID in order to find out if photographs show the same wall crack: 140 photos from 17 walls that were obtained during one visit (6 of these were used for validation and 11 for testing; 19 test identities are available for matching; $n_Q=305$). Real photos are used to account for the changes that take place during a single visit, while the features relating to appearance and growth are analysed only with the help of synthetic images. The study of cross-visit performance using real photos will be undertaken in future research. All the accuracy figures are point estimates derived from a small sample grouped by wall.

The evaluation looks at ten zero-shot matchers on CRACKID by dividing up the candidate retrieval phase from the step where a decision is made about whether or not the defects are the same and by incorporating open-set identification with rejection, since a high rank by itself is not enough to generate a report; it also makes use of the context along with crack-erased control cases to tell the cracks apart from wall-location cues, and a geometric baseline is applied in order to align the walls and then compare the projected masks.

The study looks at Research Question (RQ1), which is about association in the case of acquisition change; RQ2 is concerned with context and geometric alignment; and RQ3 deals with synthetic appearance or growth change. For RQ1 and RQ2 we use real single-visit photos, whereas RQ3 is an exploratory study based on synthetic data. Future work will focus on ticket-level duplicate reduction and cross-visit evaluation.

The contributions are as follows: (i) the cross-observation identities for CRACKID together with the annotation provenance mentioned in Section III-B; (ii) a benchmark that involves retrieval, making same-defect decisions, and open-set identification (Sections IV and VI); (iii) context controls, geometric reference, and coverage analysis (Sections V, VI-C,

and VII-B); (iv) a synthetic revisit probe (Section VII-C).

II. RELATED WORK

The analysis of cracks focuses on detection and segmentation for single images [5]–[7]; so do the consolidation studies [1], [2], and the multi-view pipelines combine one inspection into a surface or segmentation [8] without taking into account instance correspondence. Structural-health monitoring involves tracking cracks over time using fixed cameras and designated locations [9]–[12], in which the identity of the cracks is *assigned at the time of construction*: unlike in our open-set association task, they are dealing with a known crack.

The scenario involves object re-ID [4] with open-set identification [13]: Scale-Invariant Feature Transform (SIFT) [14] and Oriented FAST and Rotated BRIEF (ORB) [15], the learned correspondences such as SuperGlue [16] and LoFTR [17], and the region embeddings [3], [18]–[21] are all assessed in a zero-shot manner without CRACKID fine-tuning (the YOLOv8 row: layer-9 `yolov8n.pt` features [22]). LightGlue [23] and NetVLAD [24] are mentioned as future retrieval baselines. Vehicle-damage re-ID across claims [25] follows the same open-set protocol as ours; related work focuses on detection and severity, not on association [26]–[28]. The approach of aligning first and then reasoning about presence or absence constitutes change detection [29], [30] using planar registration [31]–[33] and an Enhanced Correlation Coefficient (ECC) fallback [34]; our key point is that the alignment and the matching targets must be *disjoint*, with the homography being derived from the wall when the crack is identified.

III. DATA ACQUISITION

A. Acquisition and composition

The CRACKID frozen benchmark snapshot consists of 140 photographs of 17 exterior walls that were taken at a single site over four half-days using two consumer handheld devices (Mendeley Data [35]; Fig. 1). The walls are either stippled render (textured) or smooth painted plaster and the only features that are visible are cracks. The dataset is divided such that six walls are used for validation (walls 01 to 06; 66 photos, 66 identities) and eleven walls for testing (walls 07 to 17; 74 photos, 146 identities). Only those identities that appear in at least two photographs are included in the answerable queries (18 validation, 19 test; $n_Q=439$ validation and 305 test queries). Walls 18 and 19 were added afterwards to the dataset and are therefore excluded from all the results. Each wall was photographed during a continuous walk-around (with a median time of 3 seconds between frames and a maximum of 81 seconds); the data include within-visit viewpoint and scale correspondences, which represents an upper bound on the performance that can be achieved with multiple visits.

For adjacent-frame retrieval, the gallery includes the adjacent frame for all 305 test queries for which an answer is required. The gap sweeps described in Sec. VII-B (involving 305 queries down to 205, then to 149, and finally to 105 for gaps 0–3) maintain Rank-1 as the number of items in the pool

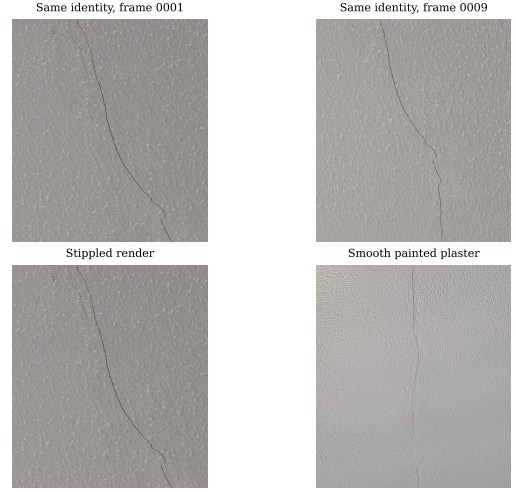


Fig. 1. Real CRACKID examples. Top: one identity across non-adjacent frames. Bottom: stippled render versus smooth plaster. Labels via homography propagation, unaudited.

decreases, which shows that the method is robust when faced with short baselines.

B. Annotation, and why it needs auditing

The ORB homography method carries each single click from frame to frame and then combines the fragments to form physical-crack identities; the merging process is the greatest risk (since an incorrect merge can turn a hard negative into a false positive). In the frozen snapshot, 95 of the 212 identities (45

IV. BENCHMARK PROTOCOL

The benchmark assesses visual association by determining, for a new crack observation, whether it matches one of the recorded defects; since annotated identities serve as a stand-in for permanent defect records, the scores indicate the accuracy of the association rather than the reduction of duplicates at the ticket level.

A. Task formulation and candidate records

A wall is one physical wall; an identity links observations of the same crack (Sec. III-B). A query is one crack instance; its gallery holds other instances from photos of the same known wall. Cross-wall search is excluded. A query is *known* if its gallery contains the identity, it is *unknown*. Each method scores query–gallery pairs for ranking, thresholded decisions, and identification with rejection. Pairs are the unit for pairs; image pairs for registration/assignment. The Hungarian one-to-one constraint applies per image pair only.

Counts for the frozen snapshot (6 validation walls 01–06; 11 test walls 07–17; walls 18–19 excluded) are in Table I. Registration covers all 301 same-wall test image pairs; 255 yield scored pairs with validity-mask cells (Sec. VII-B).

TABLE I
COUNTING UNITS PER METRIC (FROZEN SNAPSHOT, $n_Q=305$ TEST).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	305
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301
Scored image pairs (with valid cells)	341	255

B. Test conditions, decisions, and metrics

The primary test involves actual walk-arounds for each wall, assuming that the morphology remains unaltered; the only variations considered are viewpoint, distance, scale, lighting, and visibility, along with same-wall distractors and unknown queries. The validation and test walls are entirely separate. All the methods are used exactly as they are on the shared masks without any tuning of CRACKID (see Sec. V); the thresholds determined on the validation set are applied to the test set, considering only within-visit correspondence. A synthetic check is carried out to distinguish between photometric effects (condition B: blur/noise/contrast with unchanged structure) and growth effects (condition A: one tip extended) using ten photos taken from five walls with a single seed and a common warp (the parameters are available with the code), and this is done by comparing OSNet with context to HYBRID-SKEL ($\tau=8.0$ px), which is different from REG+CHAMFER: this is only a preliminary check.

Decisions: any pairs that exceed the threshold are accepted; for each query only the highest-ranking gallery entry that is above the threshold is accepted; for each image block the Hungarian assignment is used with those below the threshold being dropped. Unscorable pairs are retained (they are predicted to be negative for F_1 and are given the last rank in retrieval); a registration failure is defined as the absence of evidence, not as evidence of a new defect. The metrics used are Rank-1 and mean Average Precision (mAP); the pairwise F_1 is compared with the oracle F_1^* versus the validation $F_1^{\text{@}v}$; the assignment F_1^a is equal to $2\text{TP}/(2\text{TP}+\text{FP}+\text{FN})$; the open-set detection and identification rate (DIR) and false alarm rate (FAR) are reported with the DIR given at a FAR of 0.1; together with coverage (the fraction that is scored). The references include random ranking and the best-constant F_1 $2p/(1+p)$ (0.305 at a prevalence of 0.180; 0.544 on the 0.48 geometric subset; 0.452 in the validation at 0.292); the full chance levels are given in Table II. Since there are 11 walls and 19 matchable identities we provide the point estimates (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The baseline tests whether crack locations on the wall drive the association: it aligns wall surfaces to a common frame and then compares the projected crack masks. It assumes a roughly planar wall and uses no CRACKID training. The shared masks (trained on public crack data only; no CRACKID

wall in training) use instance extraction at 200 px and 5 px to keep inputs consistent across methods. Masks are taken as given, and their error is not measured.

A. Registration using wall texture

For I_a, I_b , a homography H ($x' \sim Hx$) is estimated from the surrounding wall texture with detected cracks dilated and excluded, so alignment is driven by surface features independently of the identity under test. This independence is conditional on the masks and the succeeding stage, not guaranteed per pair. Full detector settings (Scale-Invariant Feature Transform (SIFT) with Contrast Limited Adaptive Histogram Equalization (CLAHE) [36], ratio test, robust estimation) are archived with the code.

B. Registration cascade and validity checks

Textured render generally provides background features, whereas smooth plaster does not. A three-stage cascade is used: masked SIFT; if too few keypoints remain, unmasked SIFT as fallback (flagged per pair, since the crack then aids alignment); finally, Enhanced Correlation Coefficient (ECC) photometric alignment. Candidates pass plausibility tests (no mirrored or folded quads; scale within $[1/16, 16]$); the surviving stage is recorded per pair, and its inlier root-mean-square error (RMSE) sets the comparison tolerance. Unscored pairs count as lack of evidence, not evidence of different defects.

C. Crack projection and geometric comparison

Accepted registrations project instances from I_a into I_b for comparison by symmetric mean Chamfer distance in (1), which tolerates the small misalignments that collapse intersection over union (IoU) on narrow cracks. Let a, b be crack point sets and $d(p, b) = \min_{q \in b} \|p - q\|_2$:

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right]. \quad (1)$$

Early rejection of distant candidates applies. Chamfer coverage (fraction within tolerance τ of the per-pair RMSE, clamped to $[2, 16]$ pixels, default 6 pixels; HYBRID-SKEL uses 8.0 pixels) is evaluated separately as its scale-free counterpart (see Sec. VII-B).

D. Correspondence decisions and limits of spatial evidence

For each image pair, one-to-one assignment follows the Hungarian method [37] with the validation threshold shared by all methods; unscored pairs take value $-\infty$ (predicted negative, ranked last, as in Sec. IV). An aligned projection onto visibly empty wall is a *licensed negative*—evidence of absence, stronger than low embedding similarity—while missed detections, out-of-frame areas, and registration failures stay unresolved. The REG+CHAMFER reference targets unchanged damage; progression uses HYBRID-SKEL. Performance depends on surface-dependent failures and on homography-based annotation provenance.

TABLE II

MAIN RESULT ON CRACKID TEST (11 WALLS, FROZEN SNAPSHOT, $n_Q=305$). F_1^* USES AN ORACLE TEST THRESHOLD; ΔF_1^* IS EXCESS OVER THE FULL-POOL FLOOR; $F_1^{\text{@}v}$ USES THE VALIDATION THRESHOLD. BOLD IS THE BEST CROP PER COLUMN. GEOMETRY COVERS 0.48 WITH FLOOR 0.544, SO ITS ΔF_1^* IS NOT COMPARABLE TO CROP ROWS (SEC. VII-B). S/QUERY IS TOTAL-S/305 ON ONE WORKSTATION (RELATIVE TIMES ONLY).

Method	Family	R@1	mAP	DIR @FAR= 0.1	F_1^*	ΔF_1^*	$F_1^{\text{@}v}$	cov.	s/q
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.032	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.038	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.052	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.052	0.337	1.00	7.190
YOLOv8 backbone [22]	embedding	0.525	0.439	0.134	0.374	0.069	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.039	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.032	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.032	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.033	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	—	0.582	0.48	27.92

VI. THE APPEARANCE CEILING

A. Setup

The following ten methods employ crack-centred crops together with a common validity mask: SIFT and ORB combined with Random Sample Consensus (RANSAC) inliers, SuperGlue and LoFTR using pre-trained indoor weights, five embedding methods, and re-ID OSNet with cosine similarity, all of which are applied in a zero-shot manner on CRACKID; YOLOv8 makes use of the layer-9 `yolo8n.pt` features. The input sizes and checkpoints are available in the code. For context control, fixed backbones (OSNet, CLIP) are used: `@ctx1` increases the size of the instance box by $1\times$ in each direction; `@ctx1@nocrack` fills in the crack in order to isolate the wall context. REG+CHAMFER first aligns the walls and then compares the projected masks (as described in Sec. V); COVERAGE-ONLY continuously evaluates the registered pairs without examining the cracks. The synthetic tests compare OSNet with context against the HYBRID-SKEL variants.

B. Result: appearance buys almost nothing on the decision

Table II (11 walls, frozen snapshot, $n_Q=305$) displays a Rank-1 score ranging from 0.370 to 0.613 (with the best result being that of ViT-B/16; the highest mAP is achieved by OSNet at 0.497; and the best crop performance with DIR at a FAR of 0.1 is 0.384 for SuperGlue), although the oracle F_1^* lies between 0.337 and 0.374 and the validation-calibrated $F_1^{\text{@}v}$ is in the range 0.331 to 0.338; therefore, a high ranking does not mean that decisions have been made. The small range of the $F_1^{\text{@}v}$ values indicates a poor transfer of thresholds across the walls. Since there are 19 matchable test identities, we do not claim to have any fine-grained crop ranking. All the values given are point estimates.

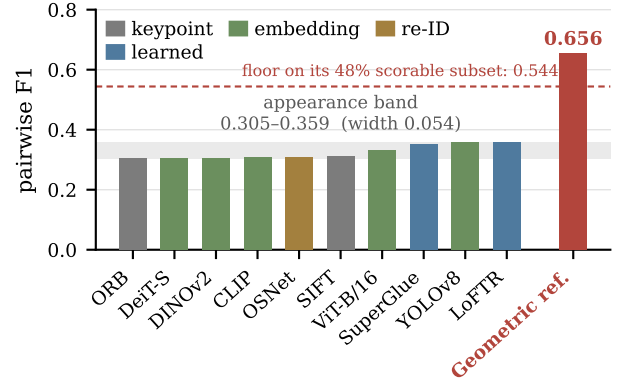


Fig. 2. Appearance ceiling: pairwise F_1^* stays narrow while ranking spreads. Geometry (red) leaves the band but uses the 0.48 pool with floor 0.544 (Table II).

C. Is it the crack, or the wall behind it?

Context contributes to both ranking and open-set performance, but the majority of the improvement is lost when the crack is removed: for OSNet the Rank-1 score rises from 0.603 to 0.775 (down to 0.711 when the crack is erased), the DIR at FAR= 0.1 goes from 0.154 to 0.404 (down to 0.393 when erased); for CLIP it goes from 0.143 to 0.311 (down to 0.286 when erased). The erased views outperform all the tight crops, suggesting that wall location is the cue. The pairwise F_1^* remains between 0.307 and 0.361 for the base, context, and erased views, although open-set performance improves by +0.250 for OSNet. Within-visit retrieval does work, but reliable automatic decisions do not; ranked lists might help the operators, though more evidence is needed regarding acceptance (Fig. 2).

TABLE III

A COMPARISON IS GIVEN BETWEEN HYBRID-SKEL AND OSNet+CTX IN THE CASE OF THE SYNTHETIC REVISIT SCENARIO, WITH THE F_1 SCORE AT THE VALIDATION THRESHOLD; THERE IS ONE SEED AND FIVE WALLS, AND THE DIFFERENCE BETWEEN HYBRID-SKEL AND REG+CHAMFER IS AS INDICATED IN TABLE II.

Protocol	Method	n_q	R@1	mAP	DIR @FAR _{0.1}	$F_1^{\text{@v}}$	F_1^a
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
	OSNet+ctx	69	0.826	0.647	0.739	0.601	0.631
B (appear.)	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
	OSNet+ctx	82	0.671	0.570	0.663	0.594	0.609

VII. RESULTS AND DISCUSSION

A. Main comparison and the open-set gap

Geometry achieves a Rank-1 score of 0.952 as compared to the best crop result of 0.613 (ViT-B/16) and has an mAP of 0.886 compared to 0.497 (OSNet), its coverage being 0.48 rather than 1.00. Once the coverage is exceeded, the ranking improvements are still obtained but the open-set gains are not: at a FAR of 0.1, DIR is 0.768 as against 0.384 (SuperGlue), 0.404 (the best OSNet+ctx1), and 0.000 for coverage (the chance level being 0.038). In the case of hairlines, crop similarities almost never reject and aligned empty projections act as valid negatives. The time cost is 7.19 seconds per query (LoFTR) as against 0.006 seconds per query (OSNet); geometry takes 27.9 seconds per query and 26.0 seconds per pair (in the case of relative only; SuperGlue and LoFTR were timed separately). With respect to RQ2, alignment enables identification beyond coverage, provided that success, surface, and provenance are taken into account. The point estimates are based on 19 identities.

B. Coverage, failures, and ablations

Crops achieve 100

C. A synthetic revisit protocol

Ten source photos taken from five walls (with one seed) are subjected to the same minor warp in either case (A) under single-tip growth or (B) due to photometric corruption, and then are matched with the actual images (procedure with code; p90 tilt 75°). HYBRID-SKEL differs from REG+CHAMFER as shown in Table II. Table III provides a comparison between HYBRID-SKEL and OSNet+ctx.

For the ranking/open-set OSNet+ctx results (0.826/0.671 and 0.739/0.663 as compared to 0.275/0.274 and 0.343/0.342), the HYBRID-SKEL method attains a higher F_1^a (0.646/0.640 compared to 0.631/0.609) and its $F_1^{\text{@v}}$ is also similar (0.569/0.574 versus 0.601/0.594); no cell corresponds to the pure homography case ($H = 0.0$)

VIII. LIMITATIONS AND FUTURE WORK

The study carries out the first measurable test of re-identifying a crack in a single visit by means of a geometric reference. The results are derived from a small number of clustered walls, labels provided by a single annotator, and

partial registration coverage. Each of the points listed here corresponds to a finding together with a step for verifying it.

- 1) **Scale to more walls and visits.** Collect data from more walls, different surface types, and additional annotators to improve the confidence intervals and the reliability of the rankings. Most importantly, include a second session using different distances and lighting conditions to establish a genuine cross-visit benchmark.
- 2) **Add provenance-free annotation.** One annotator reviewed all clicks and handled the unclaimed points; agreement was not assessed, and the geometric advantage depends on provenance. To detect the gap independent of provenance, it would be useful to create a small set of labels without using homography and to share score matrices on a per-pair basis for sensitivity, wall-clustered intervals, and threshold-free AP/AUROC.
- 3) **Close the coverage gap.** The baseline covers 48
- 4) **Learn from geometric supervision.** The baseline is an untrained reference and not a system that has been put into operation. The location benefit could be obtained by using pseudo-labels from the registration process or by fine-tuning a foundation model on homography correspondences, without needing to perform per-pair registration.
- 5) **Assess the ticket lifecycle.** Define task metrics such as the duplicate rate and the missed-new-defect rate, and also test segmentation, re-identification, and joint ticket assignment in a closed-loop simulation of real inspections.

Together, these steps will turn the within-visit reference into a multi-visit re-identification component that can be used operationally for tickets.

IX. CONCLUSION

We present CRACKID for testing crack re-ID: 140 photos from 17 walls with 212 identities in the frozen snapshot (6 validation, 11 test; walls 18–19 excluded); annotation provenance is stated (45% merged, 48% of test from merges) and unclaimed-click triage is complete (107 of 1,023 as no crack). All accuracies are point estimates without wall-level intervals (19 matchable test identities, $n_Q=305$); we assert the pattern, not the digits.

The main result is negative and of a structural nature: when decisions have to be made, ten zero-shot matchers from four families end up between +0.032 and +0.069 above the 0.305 best-constant floor (F_1^* ranging from 0.337 to 0.374), with validation-calibrated decisions lying between $F_1^{\text{@v}}$ values of 0.331 and 0.338. Retrieving from a gallery is different from deciding. The criterion is not merely the appearance of cracks: a geometric reference that captures the wall texture even when cracks are masked and involves Chamfer-matches being projected achieves a DIR at FAR=0.1 of 0.768 compared to 0.384 for the best crop and 0.404 for the best context on 48

In terms of operation, the open-set gain is a *licensed negative*—positive evidence of absence—since the crop embeddings exhibit only low similarity; it has not been proven to be

independent of homography-based annotation and must therefore be regarded as being conditioned by provenance. Two shortcomings still remain: a genuine second session is needed to make the benchmark cover a cross-visit scenario, and a second annotator is required together with a homography-free audit subset for the purpose of measuring agreement. The 48% coverage figure is in itself a shortcoming in terms of deployment because smooth plaster can be unanswerable. Lastly, the small single-seed synthetic revisit reverses the ranking mentioned in the headline in favour of OSNet+context, with the distinct HYBRID-SKEL method maintaining only a small lead in assignment- F_1 ; this represents a within-visit geometric reference and not a multi-visit ticket-backend solution.

ACKNOWLEDGMENT

This work was supported by “R2UTechnologies — modular systems” (C644876810-00000019), by “R2UTechnologies” Consortium, cofinanced by NextGenerationEU, through the Incentive System “Agendas para a Inovação Empresarial” (“Agendas for Business Innovation”), within the Recovery and Resilience Plan (PRR). The CRACKID dataset is available on Mendeley Data [35]. Code and benchmark scripts are archived with the submission and will be released on acceptance.

AI disclosure: Generative AI was used only for minor language editing. All methods, results, figures, and analyses are original, and the authors take full responsibility for their accuracy.

REFERENCES

- [1] S. Kulkarni, S. Singh, D. Balakrishnan, S. Sharma, S. Devunuri, and S. C. R. Korlapati, “CrackSeg9k: A collection and benchmark for crack segmentation datasets and frameworks,” arXiv:2208.13054 [cs.CV], 2022.
- [2] C. Benz and V. Rodehorst, “OmniCrack30k: A benchmark for crack segmentation and the reasonable effectiveness of transfer learning,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024.
- [3] K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, “Omni-scale feature learning for person re-identification,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 3702–3712.
- [4] Y. Qian, J. Barthelemy, E. Karupiah *et al.*, “Identifying re-identification challenges: Past, current and future trends,” *SN Computer Science*, vol. 5, p. 937, 2024.
- [5] Q. Zou, Y. Cao, Q. Li, Q. Mao, and S. Wang, “CrackTree: Automatic crack detection from pavement images,” *Pattern Recognition Letters*, vol. 33, no. 3, pp. 227–238, 2012.
- [6] Y. Liu, J. Yao, X. Lu, R. Xie, and L. Li, “DeepCrack: A deep hierarchical feature learning architecture for crack segmentation,” *Neurocomputing*, vol. 338, pp. 139–153, 2019.
- [7] M. Eisenbach, R. Stricker, D. Seichter, K. Amende, K. Debes, M. Sesselmann, D. Ebersbach, U. Stoeckert, and H.-M. Gross, “How to get pavement distress detection ready for deep learning? a systematic approach,” in *Int. Joint Conf. Neural Networks (IJCNN)*, 2017, pp. 2039–2047.
- [8] C. Benz and V. Rodehorst, “CrackStructures and CrackEnsembles: The power of multi-view for 2.5d crack detection,” in *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 5990–5999.
- [9] L. Parente, E. Falvo, C. Castagnetti, F. Grassi, F. Mancini, P. Rossi, and A. Capra, “Image-based monitoring of cracks: Effectiveness analysis of an open-source machine learning-assisted procedure,” *Journal of Imaging*, vol. 8, no. 2, p. 22, 2022.
- [10] S. Harb, P. Achanccaray, M. Maboudi, and M. Gerke, “Multi-temporal crack segmentation in concrete structures using deep learning approaches,” *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. X-G-2025, pp. 341–348, 2025.
- [11] X. Sun and P. Chang, “Robust perspective correction for real-world crack evolution tracking in image-based structural health monitoring,” arXiv:2506.22437 [cs.CV], 2025.
- [12] NEC Corporation, “Photographing guide device,” US Patent 12,223,639 B2 (priority 2019), 2025.
- [13] S. Liao, Z. Mo, Y. Hu, and S. Z. Li, “Open-set person re-identification,” arXiv:1408.0872 [cs.CV], 2014.
- [14] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.
- [15] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, “ORB: An efficient alternative to SIFT or SURF,” in *IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 2564–2571.
- [16] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, “SuperGlue: Learning feature matching with graph neural networks,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4938–4947.
- [17] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, “LoFTR: Detector-free local feature matching with transformers,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 8922–8931.
- [18] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations (ICLR)*, 2021.
- [19] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, “Training data-efficient image transformers and distillation through attention,” in *International Conference on Machine Learning (ICML)*, 2021, pp. 10 347–10 357.
- [20] A. Radford, J. W. Kim, C. Hallacy *et al.*, “Learning transferable visual models from natural language supervision,” in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.
- [21] M. Oquab, T. Darcet, T. Moutakanni *et al.*, “DINOv2: Learning robust visual features without supervision,” *Transactions on Machine Learning Research*, 2024.
- [22] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” Available: <https://github.com/ultralytics/ultralytics>, 2023.
- [23] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, “LightGlue: Local feature matching at light speed,” in *IEEE/CVF Int. Conference on Computer Vision (ICCV)*, 2023, pp. 17 627–17 638.
- [24] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, “NetVLAD: CNN architecture for weakly supervised place recognition,” in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5297–5307.
- [25] L. Maiano, A. Montuschi, M. Caserio *et al.*, “A deep-learning-based antifraud system for car-insurance claims,” *Expert Systems with Applications*, vol. 231, p. 120644, 2023.
- [26] M. J. Hasan, C. K. Nguyen, Y. L. Boo, H. Jahani, and K.-L. Ong, “Vehicle damage detection using artificial intelligence: A systematic literature review,” *WIREs Data Mining and Knowledge Discovery*, vol. 15, p. e70027, 2025.
- [27] S. A. Pérez-Zarate, D. Corzo-García, J. L. Pro-Martín *et al.*, “Automated car damage assessment using computer vision: Insurance company use case,” *Applied Sciences*, vol. 14, no. 20, p. 9560, 2024.
- [28] D. Mallios, L. Xiaofei, N. McLaughlin, J. M. Del Rincon, C. Galbraith, and R. Garland, “Vehicle damage severity estimation for insurance operations using in-the-wild mobile images,” *IEEE Access*, vol. 11, pp. 78 644–78 655, 2023.
- [29] K. Sakurada and T. Okatani, “Change detection from a street image pair using CNN features and superpixel segmentation,” in *British Machine Vision Conference (BMVC)*, 2015.
- [30] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, “Street-view change detection with deconvolutional networks,” *Autonomous Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.
- [31] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
- [32] M. A. Fischler and R. C. Bolles, “Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography,” in *Readings in Computer Vision*. Morgan Kaufmann, 1987, pp. 726–740.

- [33] D. Barath, J. Matas, and J. Noskova, "MAGSAC: Marginalizing sample consensus," in *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 10 197–10 205.
- [34] G. D. Evangelidis and E. Z. Psarakis, "Parametric image alignment using enhanced correlation coefficient maximization," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 30, no. 10, pp. 1858–1865, 2008.
- [35] D. Ferramenta, "CrackID," Mendeley Data, Version 1, Available: <https://doi.org/10.17632/t749fc4x44.1>, 2026.
- [36] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*. Academic Press, 1994, pp. 474–485.
- [37] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.